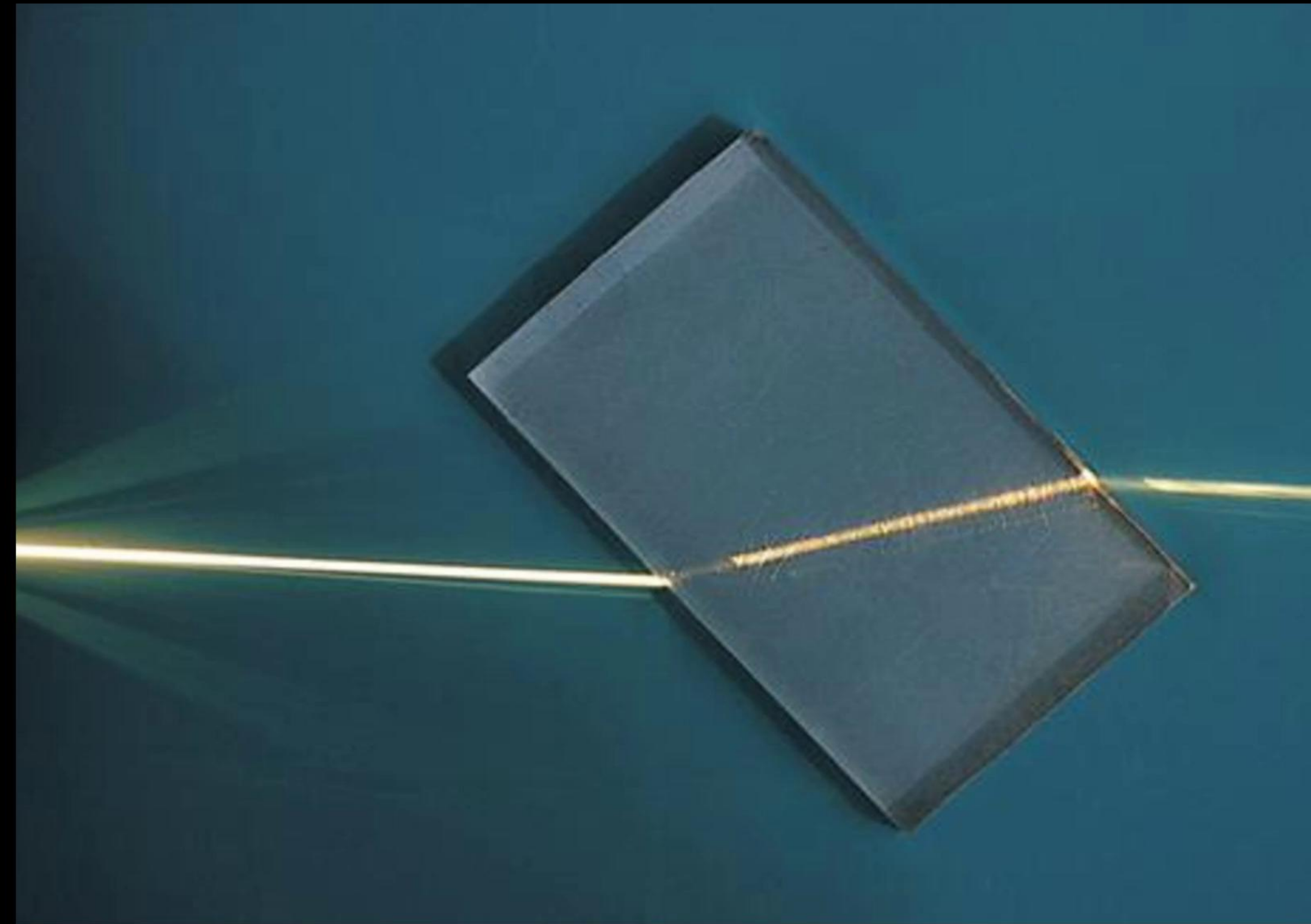


Refraction of Light

- When traveling obliquely from one medium to another, the direction of propagation of light in the second medium changes. This phenomenon is known as refraction of light.

OR

- The bending of light when it enters obliquely from one transparent medium to another.
- The major cause of refraction to occur is the change in the speed of light in different mediums
- Speed of light is maximum in vacuum : $c = 3 \times 10^8$ m/s
- **Fermat's Principle** : A ray of light going from point A to point B always takes the route of least time.



Refraction of Light

Some Examples of Refraction :

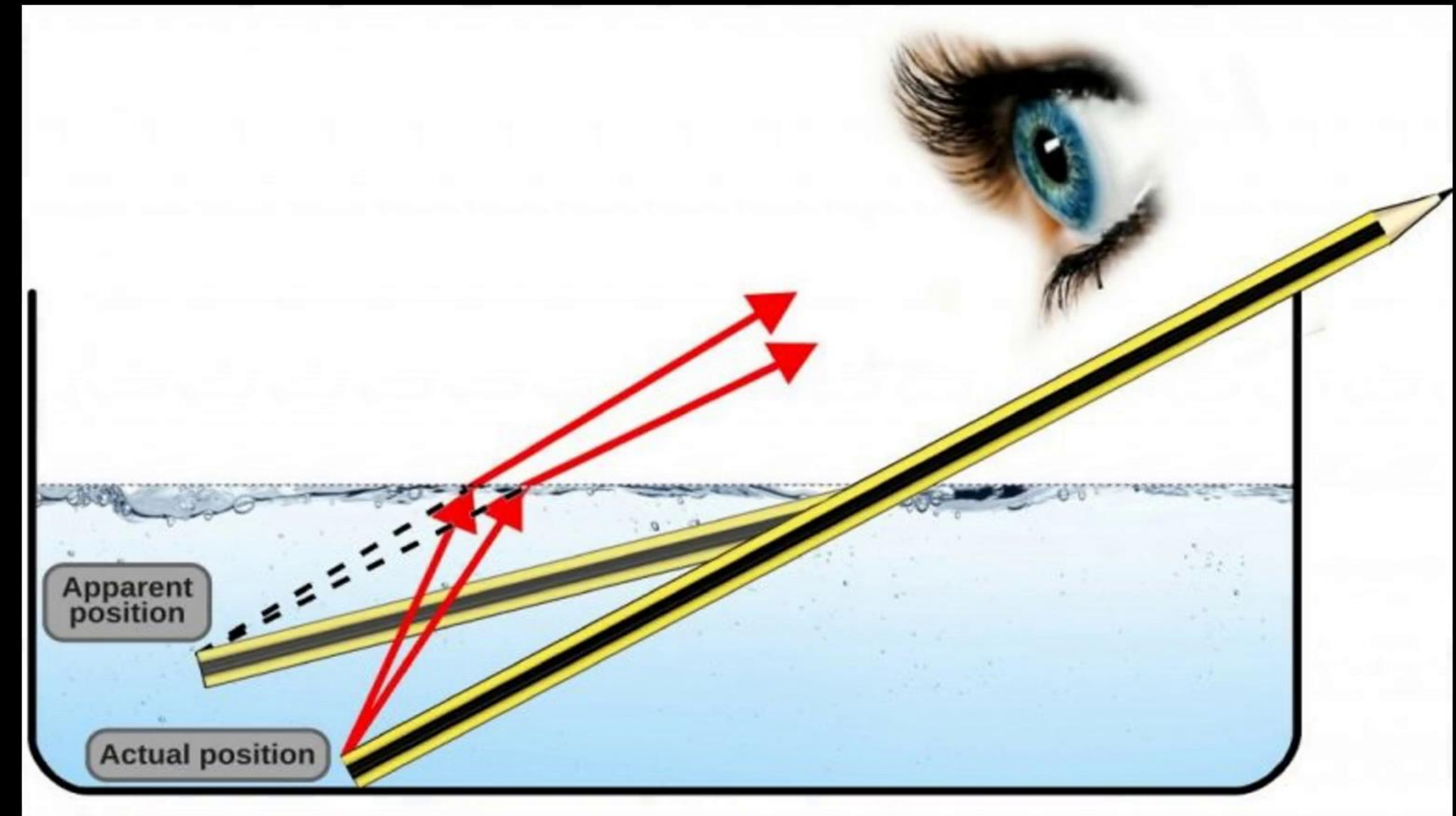
- (1) The bottom of swimming pool appears higher



Refraction of Light

Some Examples of Refraction :

- (1) The bottom of swimming pool appears higher
- (2) The pencil partially immersed in water appears to be bent at the interface



Refraction of Light

Some Examples of Refraction :

- (1) The bottom of swimming pool appears higher
- (2) The pencil partially immersed in water appears to be bent at the interface
- (3) Lemons placed in a glass tumbler appears bigger



Refraction of Light

Some Examples of Refraction :

- (1) The bottom of swimming pool appears higher
- (2) The pencil partially immersed in water appears to be bent at the interface
- (3) Lemons placed in a glass tumbler appears bigger
- (4) Letters of a book appear to be raised when seen through a glass slab.

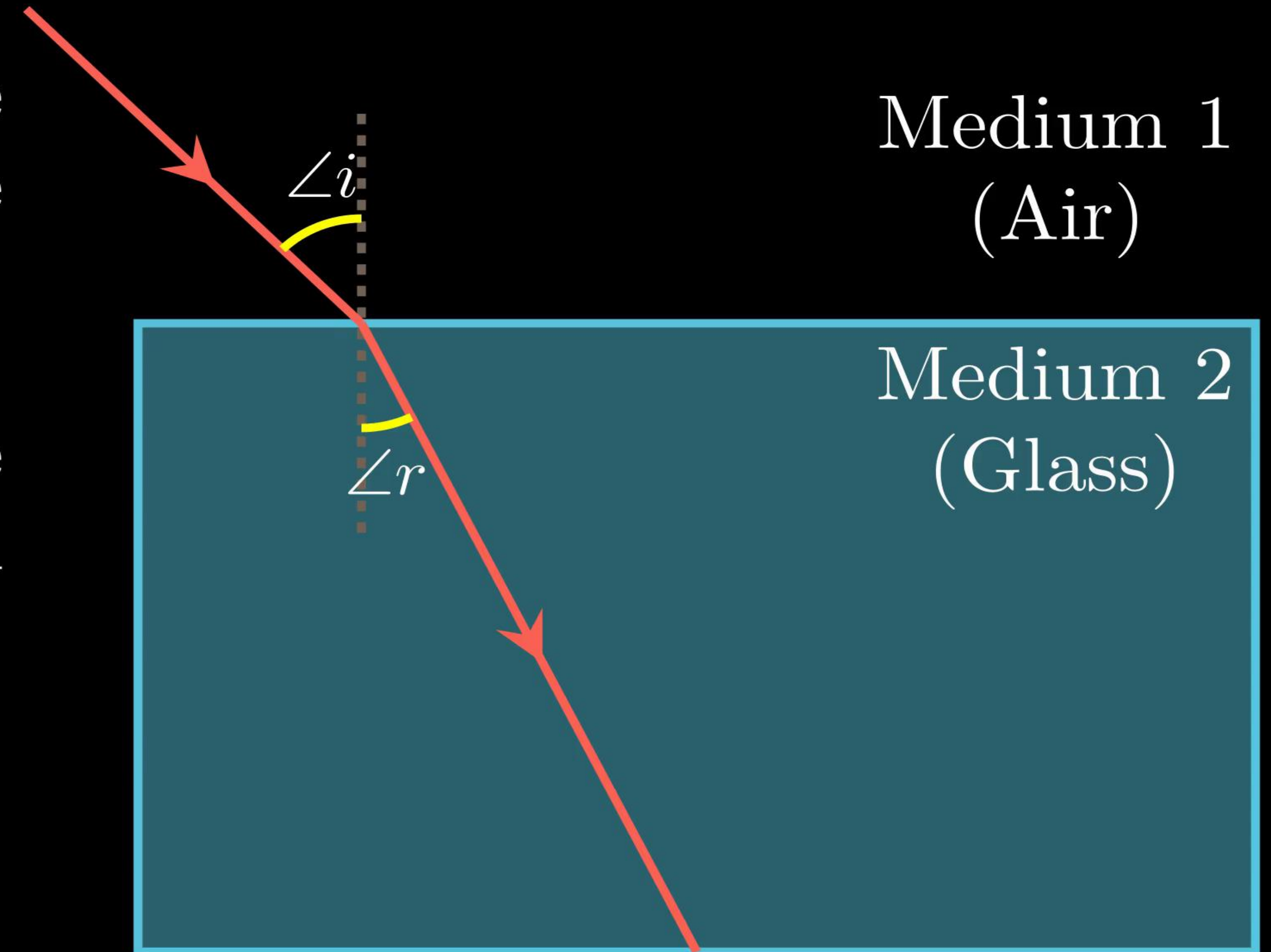


Laws of Refraction

- (1) The incident ray, the refracted ray and the normal at the point of incidence all lie in the same plane.
- (2) **Snell's law** : The ratio of the sine of the angle of incidence to the sine of the angle of refraction is a constant.

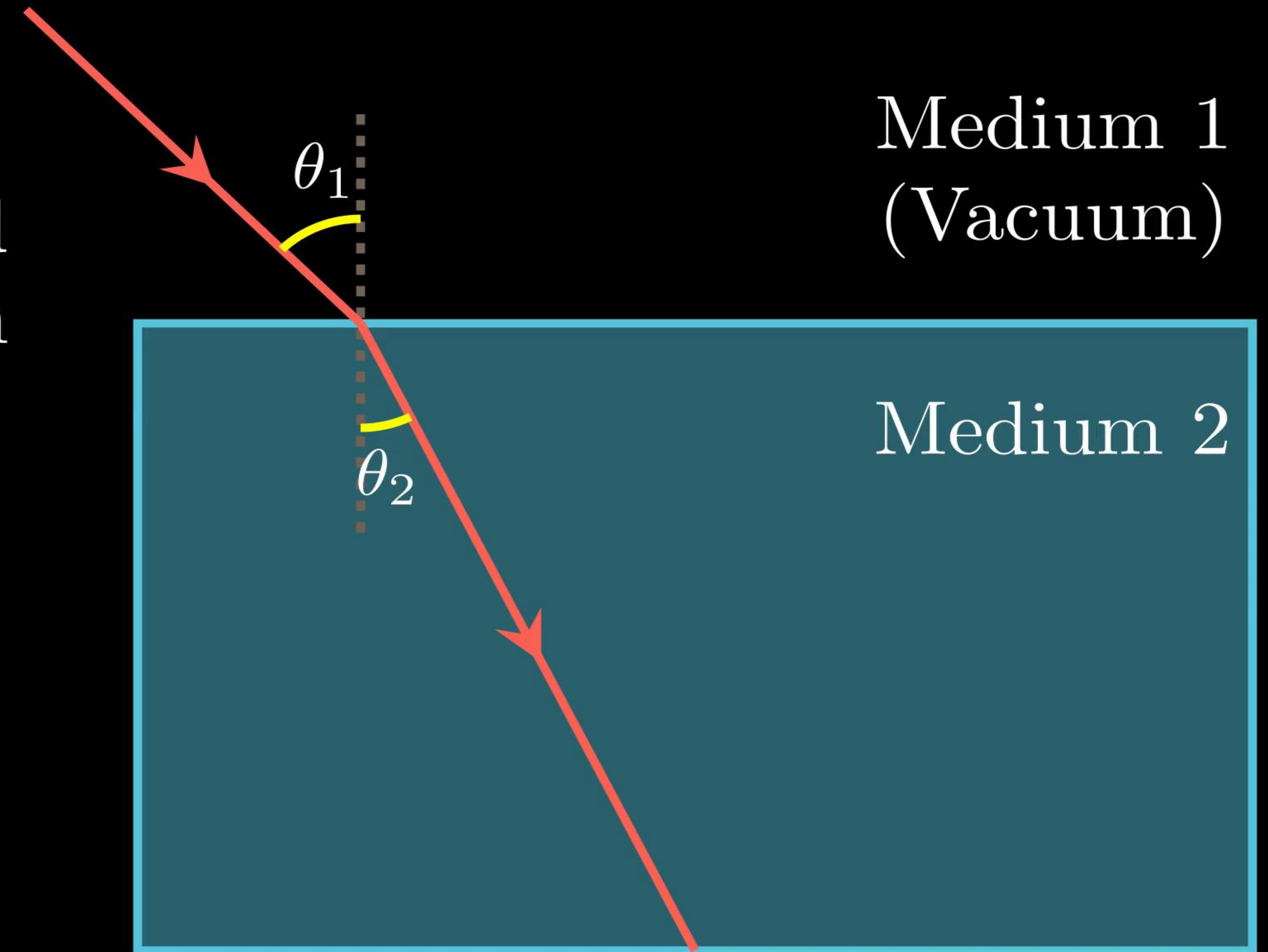
$$\frac{\sin(\angle i)}{\sin(\angle r)} = \text{Constant} = n_{21}$$

This **constant** value is called the **refractive index** (n_{21}) of the second medium with respect to the first.



The Refractive Index

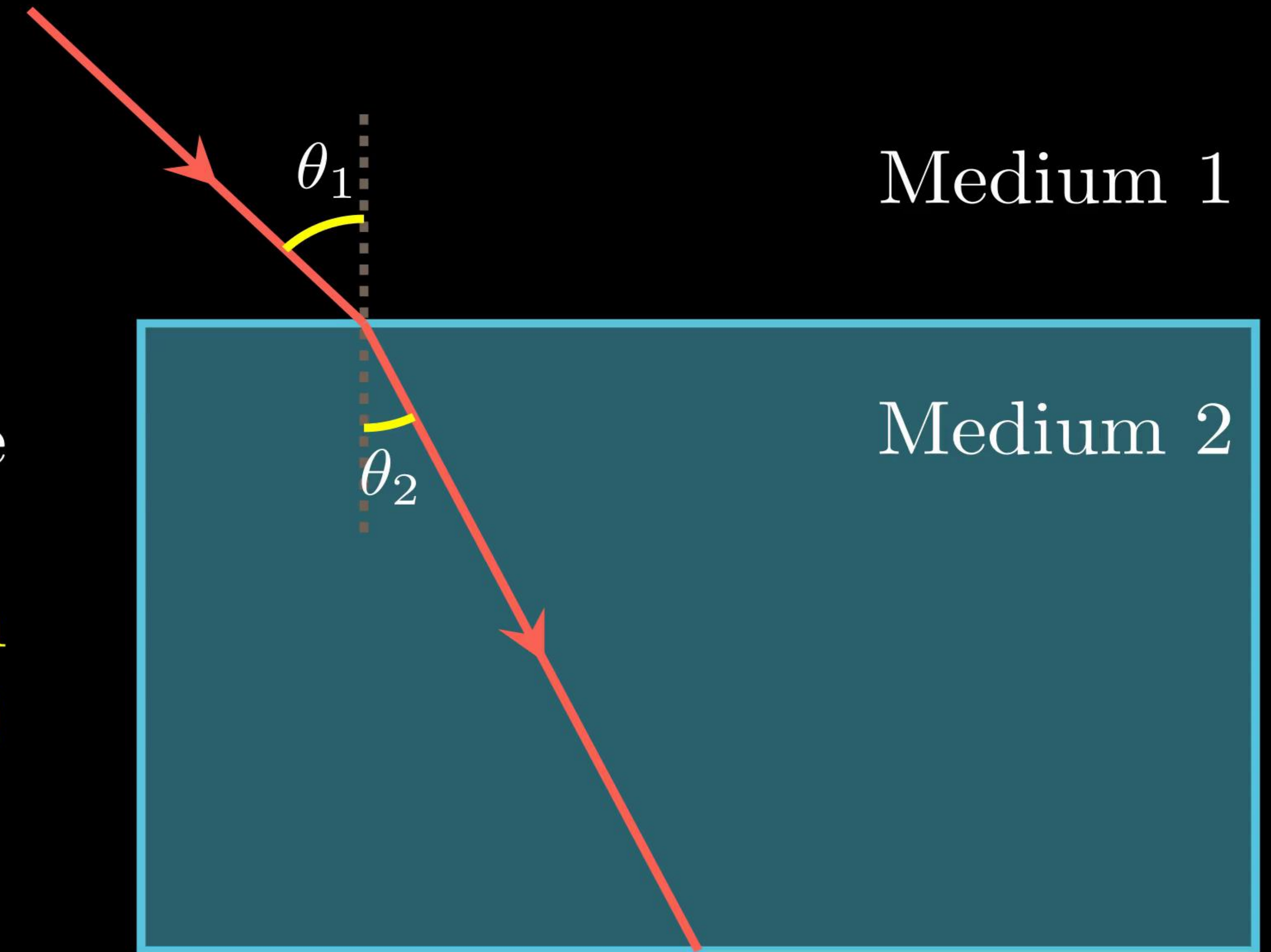
- Different medium have different abilities to bend or refract light. This bending ability of a medium is known as the refractive index.
- Types of Refractive index :
 - (1) Absolute refractive index (n)
 - (2) Relative Refractive index (n_{21})
- (1) **Absolute Refractive index** : Absolute refractive index or simply **refractive index of Medium 2** (n_2) is defined as the ratio of speed of light in vacuum (c) (i.e., Medium 1) to the speed of light in the Medium 2 (v_2)



$$n_2 = \frac{\text{speed of light in vacuum}}{\text{speed of light in medium 2}} = \frac{c}{v_2} = \frac{\lambda_0}{\lambda_2} = \sqrt{\mu_r \epsilon_r}$$

The Refractive Index

- **(2) Relative Refractive index** : Relative refractive index of medium 2 with respect to medium 1 (n_{21}) is defined as the ratio of speed of light in medium 1 (v_1) to the speed of light in the Medium 2 (v_2)



$$n_{21} = \frac{\text{speed of light in medium 1}}{\text{speed of light in medium 2}} = \frac{v_1}{v_2} = \frac{\lambda_1}{\lambda_2} = \frac{n_2}{n_1} = \frac{\sin(\theta_1)}{\sin(\theta_2)}$$

The Refractive Index

Material Medium	Refractive Index	Material Medium	Refractive Index
Air	1.0003	Crown glass	1.52
Ice	1.31	Canada Balsam	1.53
Water	1.33	Rock salt	1.54
Alcohol	1.36	Carbon disulphide	1.63
Kerosene	1.44	Dense flint glass	1.65
Fused quartz	1.46	Ruby	1.71
Turpentine oil	1.47	Sapphire	1.77
Benzene	1.50	Diamond	2.42

Table: Absolute refractive index of some material media

The Refractive Index

- **Optical Density** : The ability of a medium to refract light is also expressed in terms of its optical density.
- Optical density is not the same as mass density.
- **Denser Medium** : In comparing two media, the one with the **larger refractive index** is optically denser medium than the other.
- **Rarer Medium** : The other medium of **lower refractive index** is optically rarer.
- The speed of light is higher in a rarer medium than a denser medium
- Thus, a ray of light traveling from a rarer medium to a denser medium slows down and bends **towards the normal**.
- When it travels from a denser medium to a rarer medium, it speeds up and bends **away from the normal**.

Refraction through a Rectangular Glass Slab (Lateral Shift)

Using Snell's law -

$$n_a \sin(i) = n_g \sin(r) \quad \text{--- (1)}$$

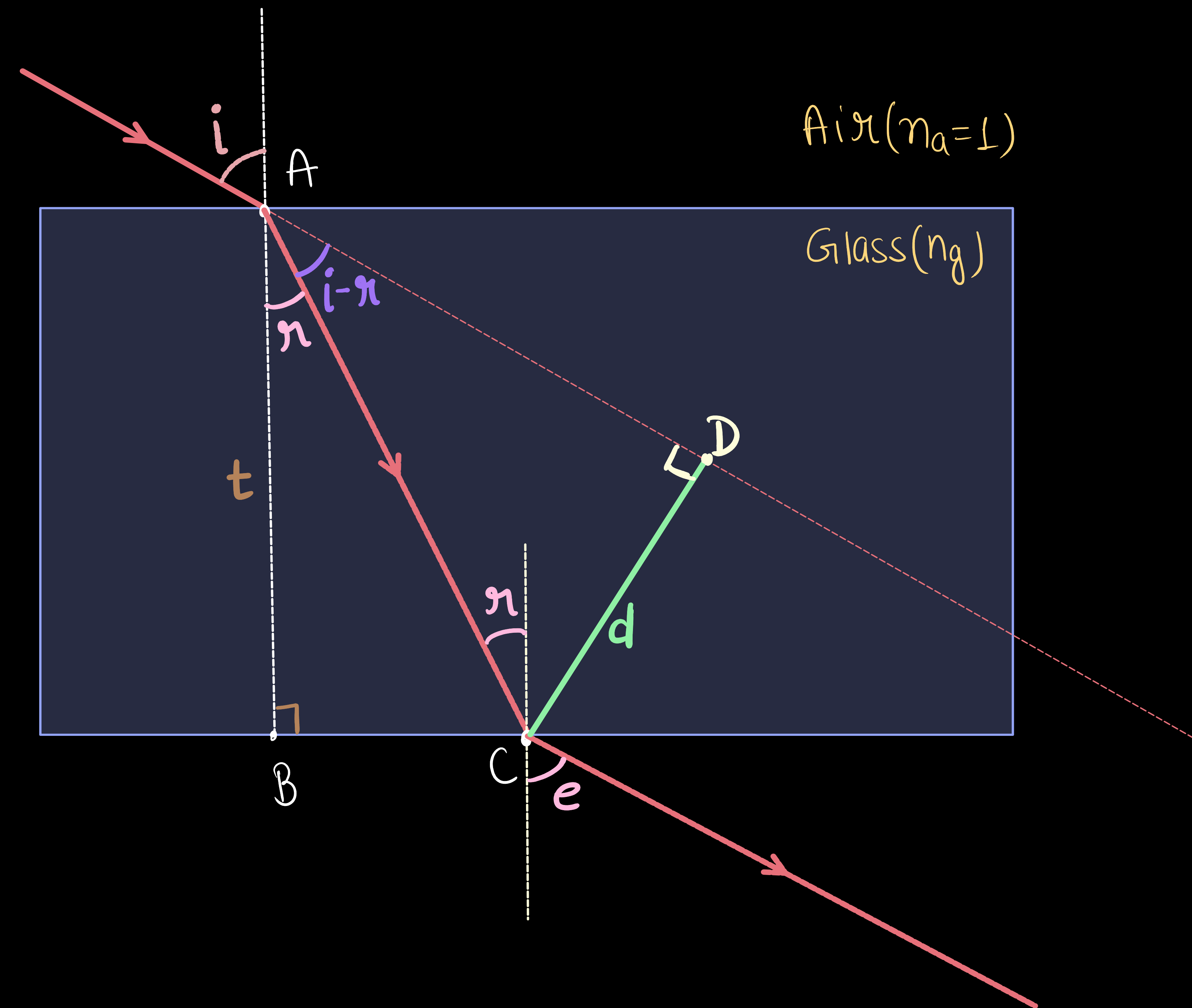
Also, $n_g \sin(r) = n_a \sin(e) \quad \text{--- (2)}$

from eqⁿ (1) & (2)

$$n_a \sin(e) = n_a \sin(i)$$

$$\sin(e) = \sin(i)$$

$$\Rightarrow \boxed{e = i}$$



Refraction through a Rectangular Glass Slab (Lateral Shift)

In $\triangle ABC$

$$\cos(r) = \frac{AB}{AC}$$

$$AC = \frac{AB}{\cos(r)}$$

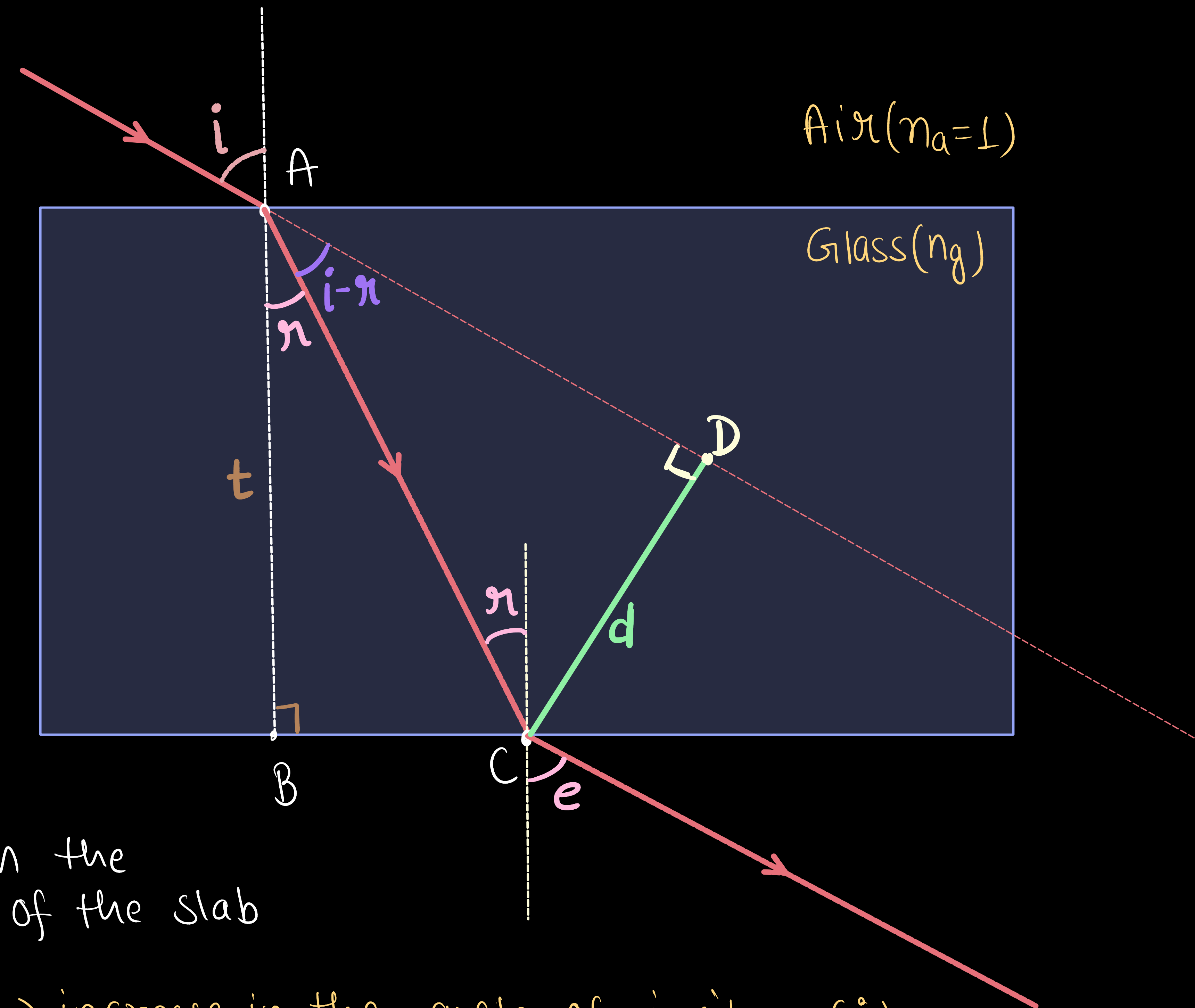
$$AC = \frac{t}{\cos(r)} \quad \text{--- ①}$$

In $\triangle ADC$

$$\sin(i-r) = \frac{CD}{AC}$$

$$CD = AC \sin(i-r)$$

$$d = \frac{t \sin(i-r)}{\cos(r)} \quad \text{--- ②}$$



increase in the thickness of the slab
 Lateral shift (d) $\rightarrow$ increase in the angle of incidence (i)
 increases with $\rightarrow$ increase in the refractive index of the slab.

Refraction through a Rectangular Glass Slab (Lateral Shift)

$$d = \frac{t \sin(i-r)}{\cos(r)} \quad \text{--- (2)}$$

$$d = \frac{t (\sin i \cos r - \cos i \sin r)}{\cos r}$$

$$d = t \left(\sin i - \cos i \frac{\sin r}{\cos r} \right) \quad \text{--- (3)}$$

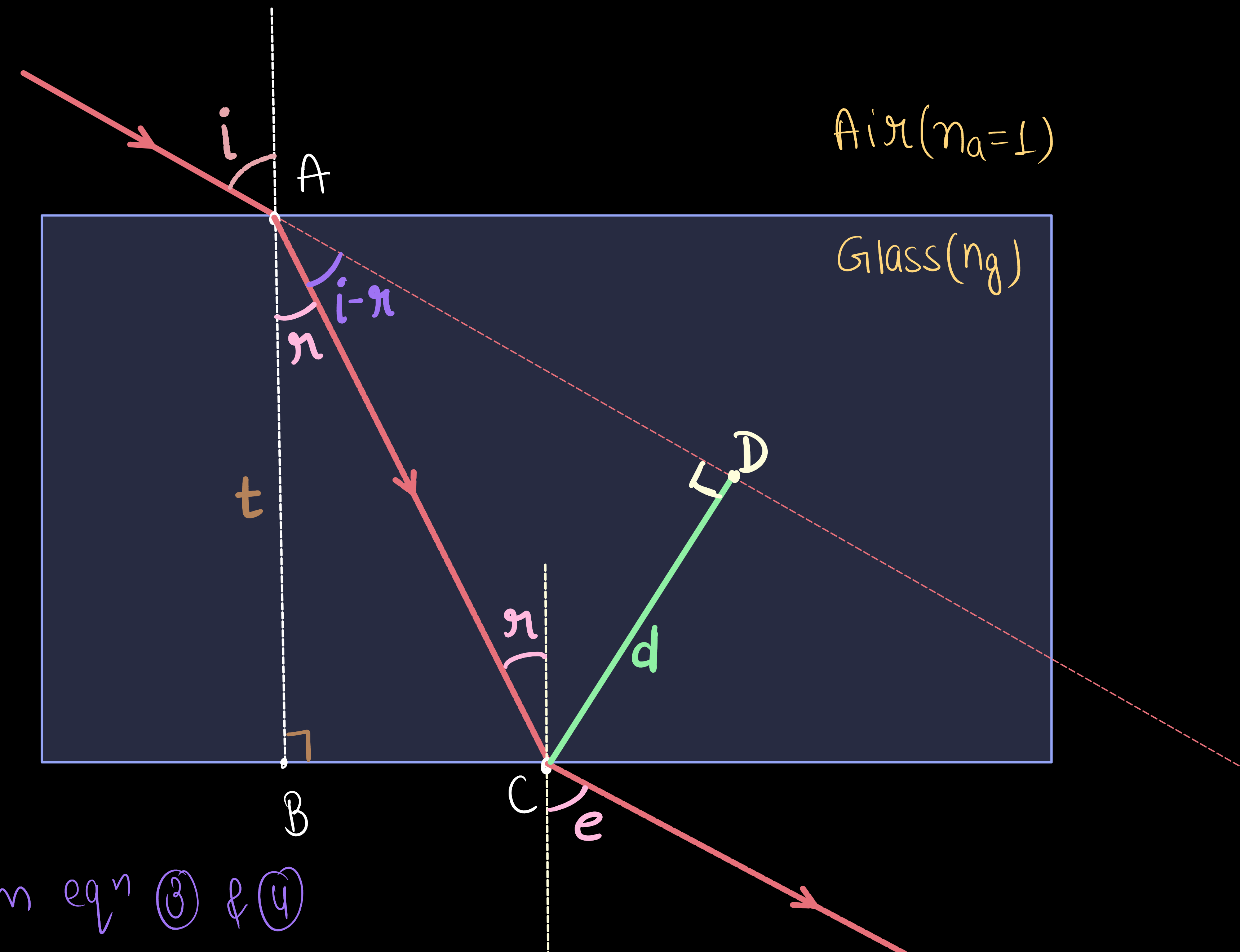
from snell's law $\rightarrow 1 \times \sin i = n_g \sin r$

$$\sin r = \frac{\sin i}{n_g}$$

$$\cos r = \sqrt{1 - \sin^2 r}$$

$$\cos r = \sqrt{1 - \frac{\sin^2 i}{n_g^2}} = \frac{\sqrt{n_g^2 - \sin^2 i}}{n_g}$$

$$\frac{\sin r}{\cos r} = \frac{\sin i}{\sqrt{n_g^2 - \sin^2 i}} \quad \text{--- (4)}$$



from eqn (3) & (4)

$$d = t \sin i \left(1 - \frac{\cos i}{\sqrt{n_g^2 - \sin^2 i}} \right) \quad \text{--- (6)}$$

Real and Apparent Depths (Normal Shift)

using snell's law -

$$n_1 \sin \theta_1 = n_2 \sin \theta_2$$

$$\frac{\sin \theta_1}{\sin \theta_2} = \frac{n_2}{n_1} = n_{21} \quad \text{--- (1)}$$

In $\triangle OAB$ and $\triangle IAB$

$$\sin \theta_1 = \frac{AB}{BI} \quad - (2)$$

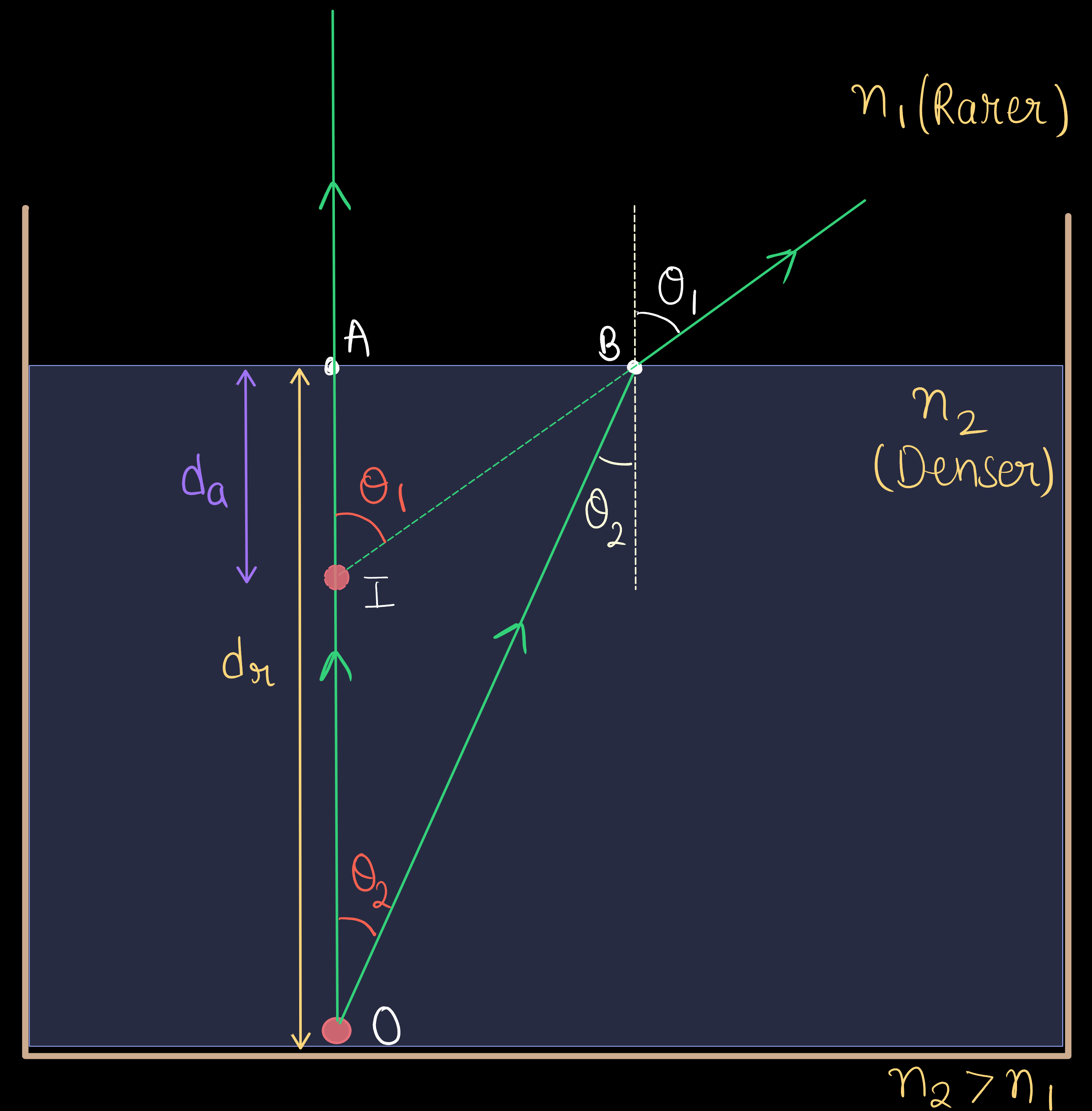
$$\sin \theta_2 = \frac{AB}{BO} \quad \text{--- (3)}$$

Dividing eq (2) & (3)

$$\frac{\sin \theta_1}{\sin \theta_2} = \frac{B_0}{B_I}$$

Since, Point B is very close to the point A

$$\frac{\sin \theta_1}{\sin \theta_2} \approx \frac{AO}{AI} \quad \text{--- (4)}$$



Real and Apparent Depths (Normal Shift)

from eqⁿ (1) & (4) -

$$\frac{AO}{AI} = \frac{n_2}{n_1}$$

$$\frac{dx}{da} = \frac{n_2}{n_1} = n_{21}$$

Derivation is not important,
only formula is important
for numericals

If first medium is air (i.e., $n_1 = 1$)

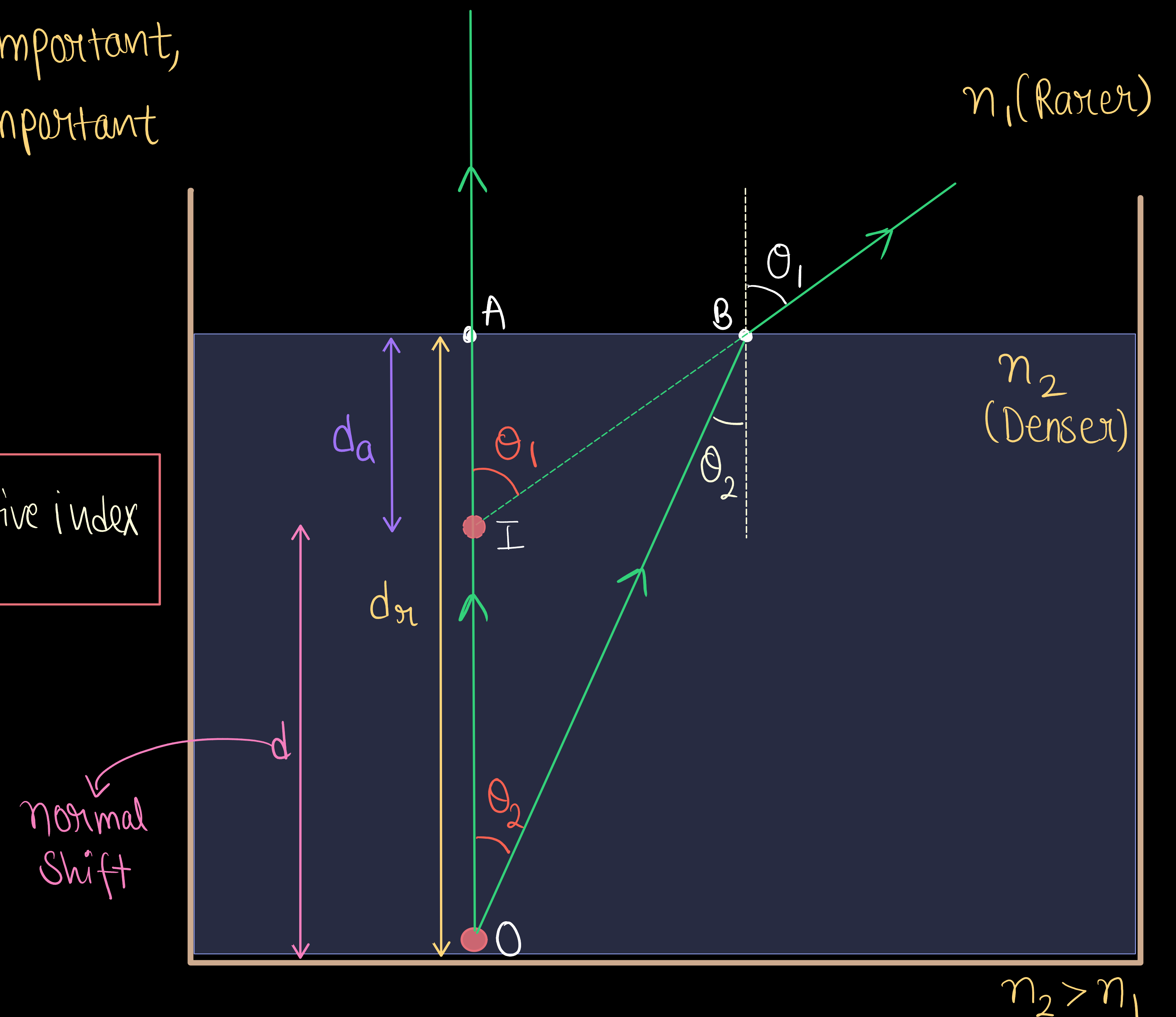
$$\frac{dx}{da} = n_2$$

$$\frac{\text{real depth}}{\text{Apparent depth}} = \text{Refractive index}$$

* Normal Shift (d) = $dx - da$

$$d = dx - \frac{dx}{n_{21}}$$

$$d = dx \left(1 - \frac{1}{n_{21}} \right)$$



Total Internal Reflection (TIR)

- When light travels from an **optically denser medium** to a **rarer medium**, a part is **reflected back** and the rest is **refracted**.

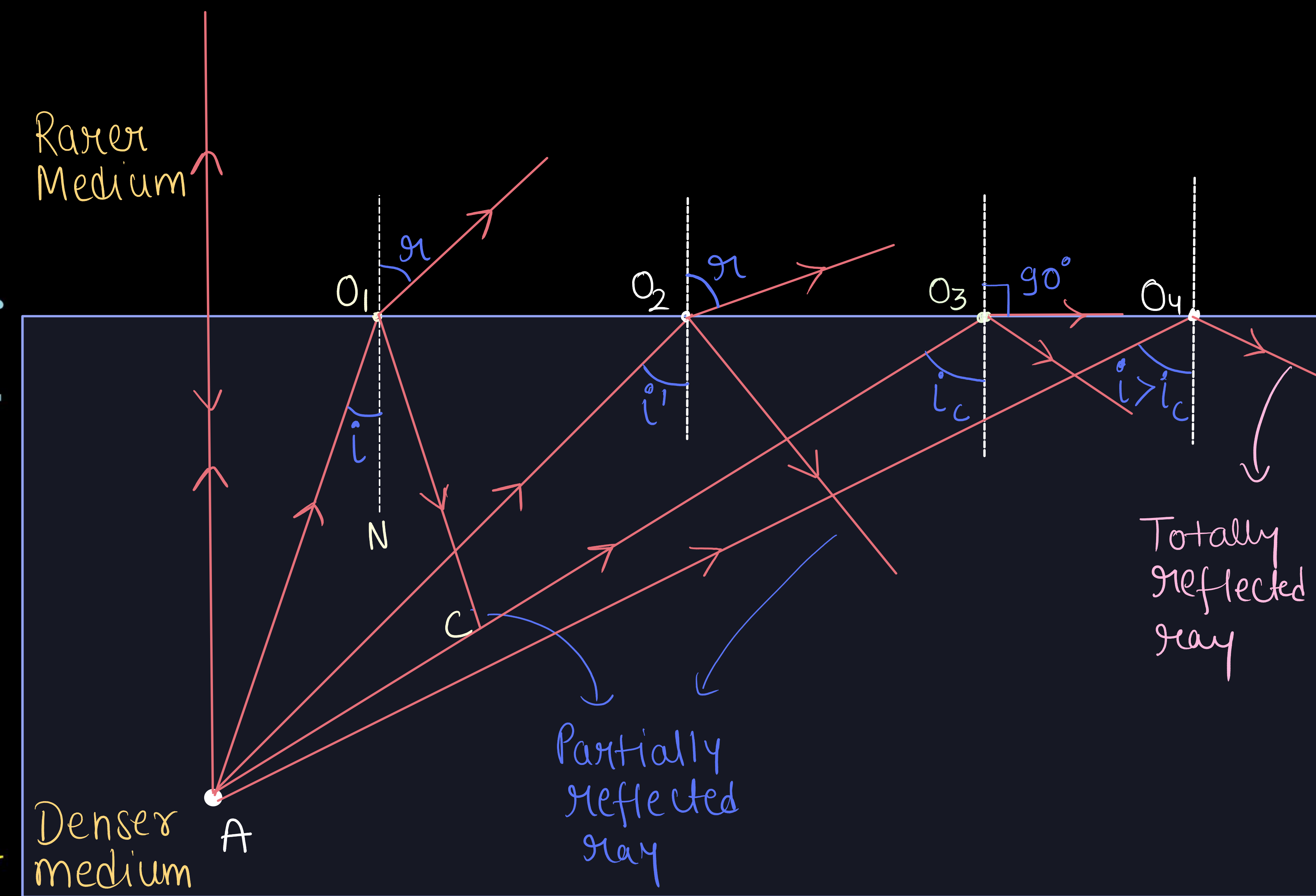
- As the **angle of incidence** (i) increases:

Angle of refraction ($r \geq i$) also increase.

At critical angle ($i = i_c$), the refracted ray **grazes the surface** ($r = 90^\circ$).

For $i > i_c$, all light is reflected (no refracted ray) — **Total Internal Reflection**.

- **Definition:** When a ray going from a denser to a rarer medium is completely reflected back into the denser medium, the phenomenon is called **Total Internal Reflection**.



Total Internal Reflection (TIR)

- Conditions for Total Internal Reflection:

1. Light must travel from a **denser to a rarer medium**.
2. The angle of incidence must be **greater than the critical angle** ($i > i_c$).

- Relation between refractive index and critical angle:

From Snell's law: $n_1 \sin i = n_2 \sin r$

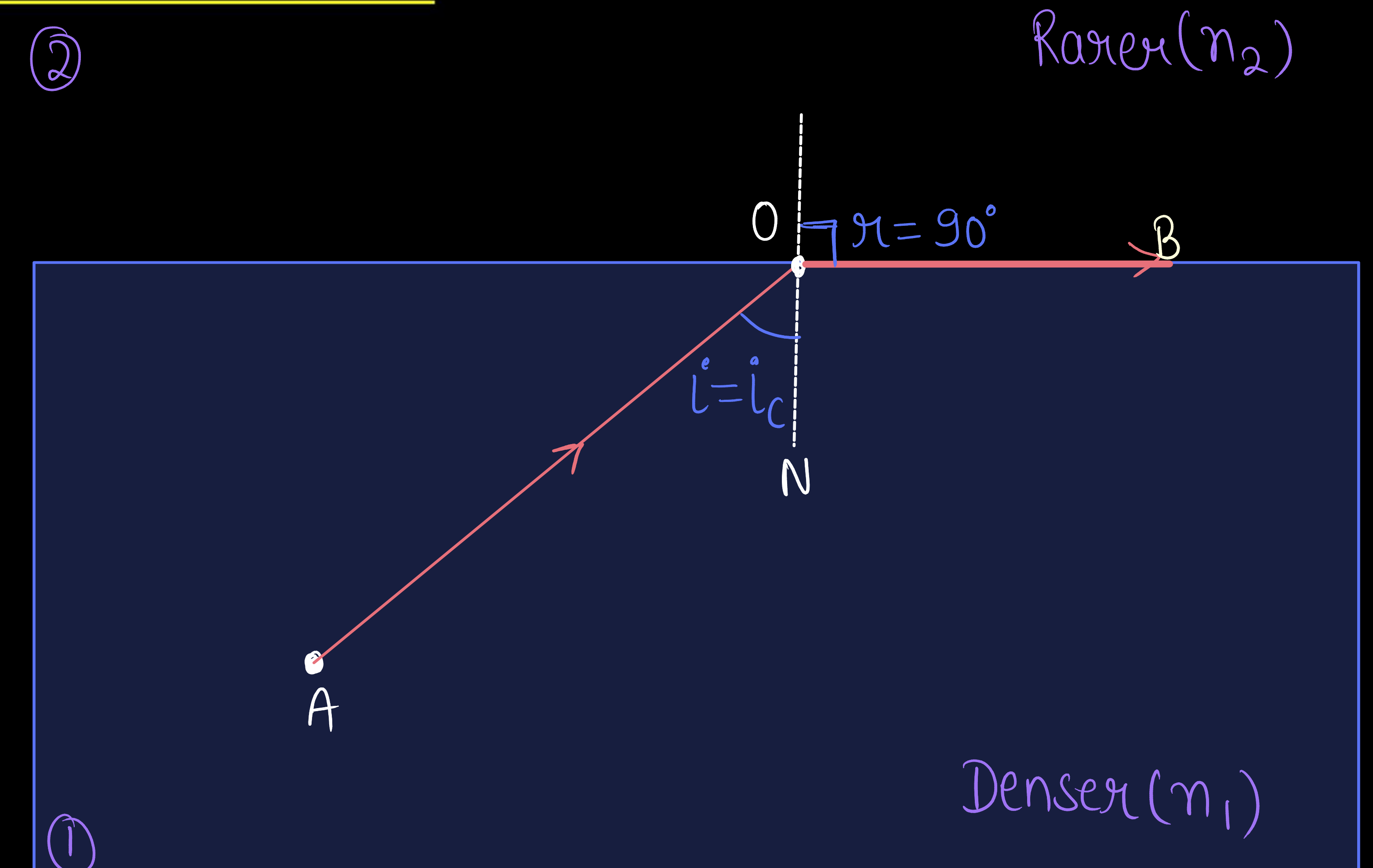
At $i = i_c$ and $r = 90^\circ$:

$$n_1 \sin i_c = n_2 \sin(90^\circ)$$

$$\Rightarrow \boxed{\sin i_c = \frac{n_2}{n_1}}$$

For rarer medium as air:

$$\boxed{\sin i_c = \frac{1}{n}}$$



Key Points to Remember (For Boards)

- **Definition:** TIR occurs when light from a denser to rarer medium is fully reflected back.
- **Conditions:** (i) Denser $\rightarrow$ rarer (ii) $i > i_c$
- **Formula:** $\sin i_c = \frac{1}{n}$ (for air as rarer medium)
- **Laws used:** Snell's law — $n_1 \sin i = n_2 \sin r$
- **Typical i_c :** Water-air 48.5° , Glass-air 41.8° , Diamond-air 24.4°
- **Main uses:** Optical fibres, prisms in binoculars, diamond brilliance, mirage.
- **Reason:** Beyond i_c , $\sin r > 1$ (impossible) $\rightarrow$ full reflection.
- **Tip:** Smaller $i_c \rightarrow$ more reflections $\rightarrow$ greater brightness.

Totally Reflecting Prism

- A **right-angled isosceles prism** (angles 90° , 45° , 45°) can be used to obtain **total internal reflection**.

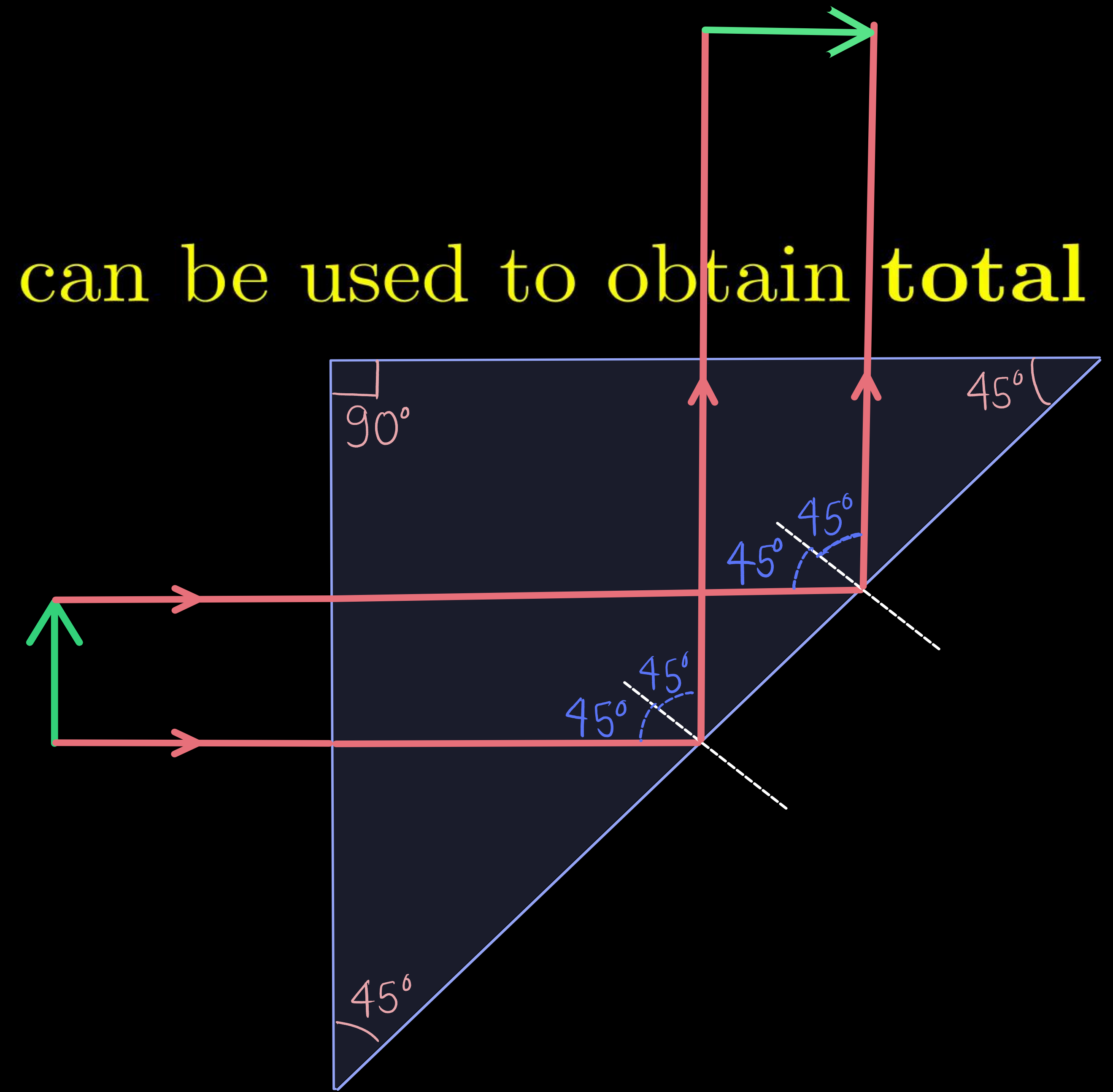
A light ray entering **normally** on one perpendicular face passes **undeviated** into the prism.

It strikes the **hypotenuse face** at an angle of 45° .

For glass-air interface, **critical angle** $i_c = 41.8^\circ$.
Since $45^\circ > i_c$, the ray undergoes **TIR**.

The reflected ray comes out **normally** from the second perpendicular face.

The prism thus **deviates the ray by 90°** .



Totally Reflecting Prism

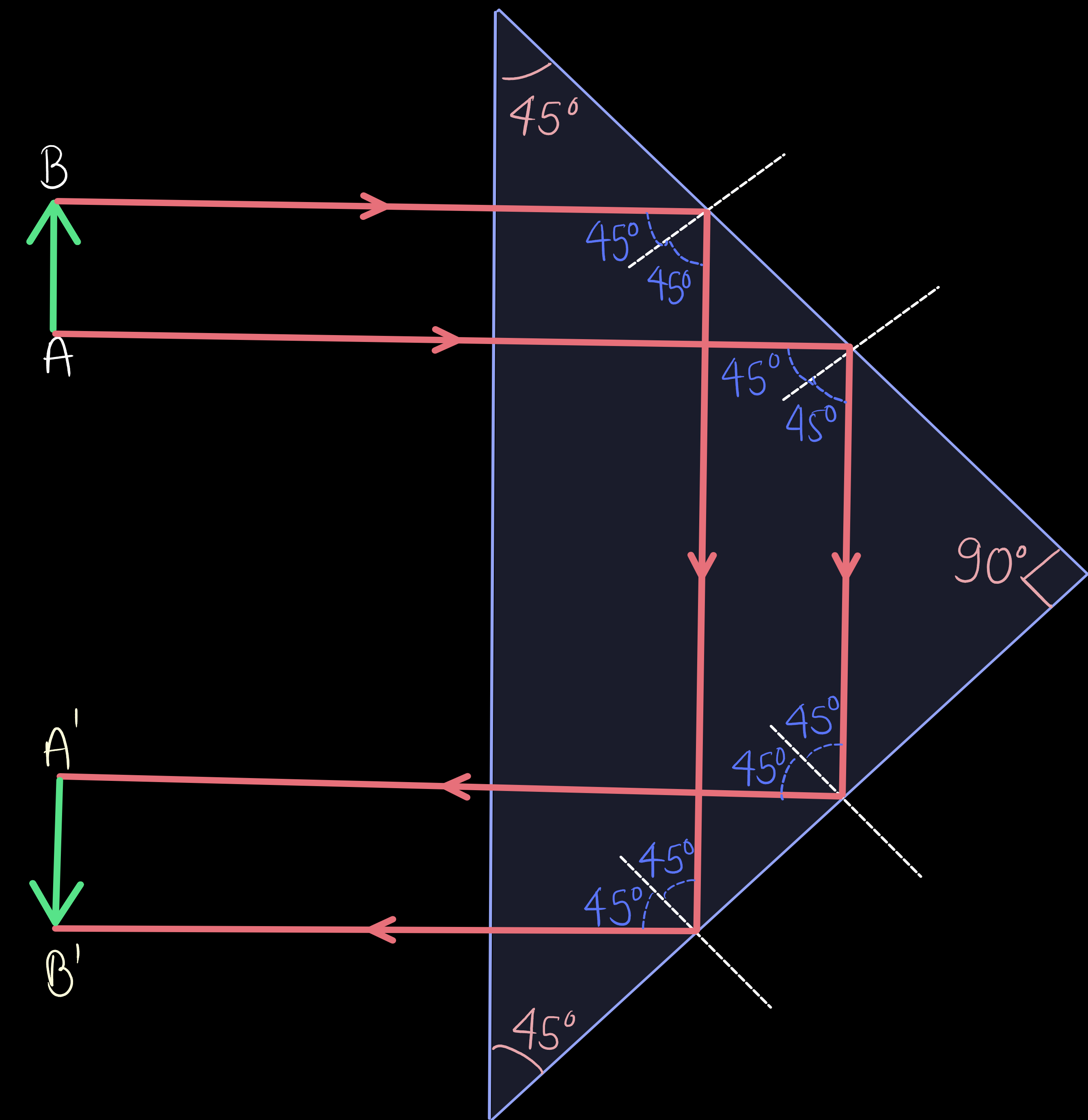
Similarly A light ray entering **normally** on hypotenuse face passes **undeviated** into the prism.

It strikes the **perpendicular face** at an angle of 45° . Since $45^\circ > i_c$, the ray undergoes **TIR**.

For glass-air interface, **critical angle** $i_c = 41.8^\circ$. Since $45^\circ > i_c$, the ray undergoes **TIR**.

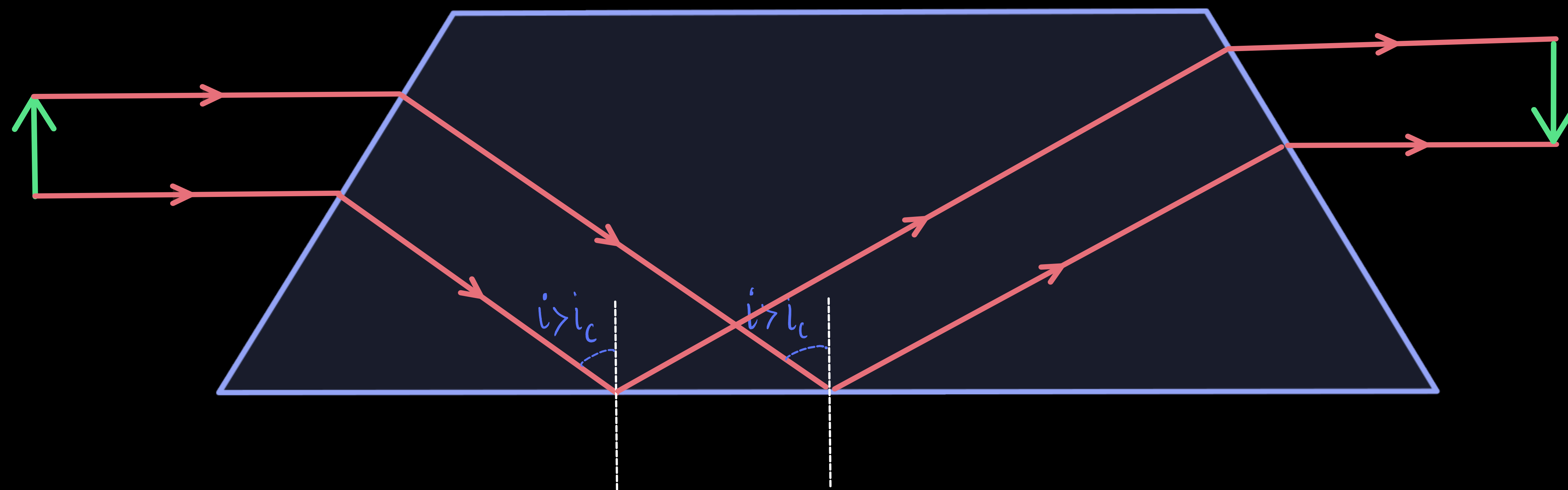
The reflected ray strikes the second **perpendicular face** at an angle of 45° and again undergoes **TIR**.

The prism thus **deviates the ray by 180°** .



Totally Reflecting Prism

- Advantages: Permanent reflecting surface; no absorption or scattering losses.
- Uses: In periscopes, binoculars, and optical instruments to change light path.

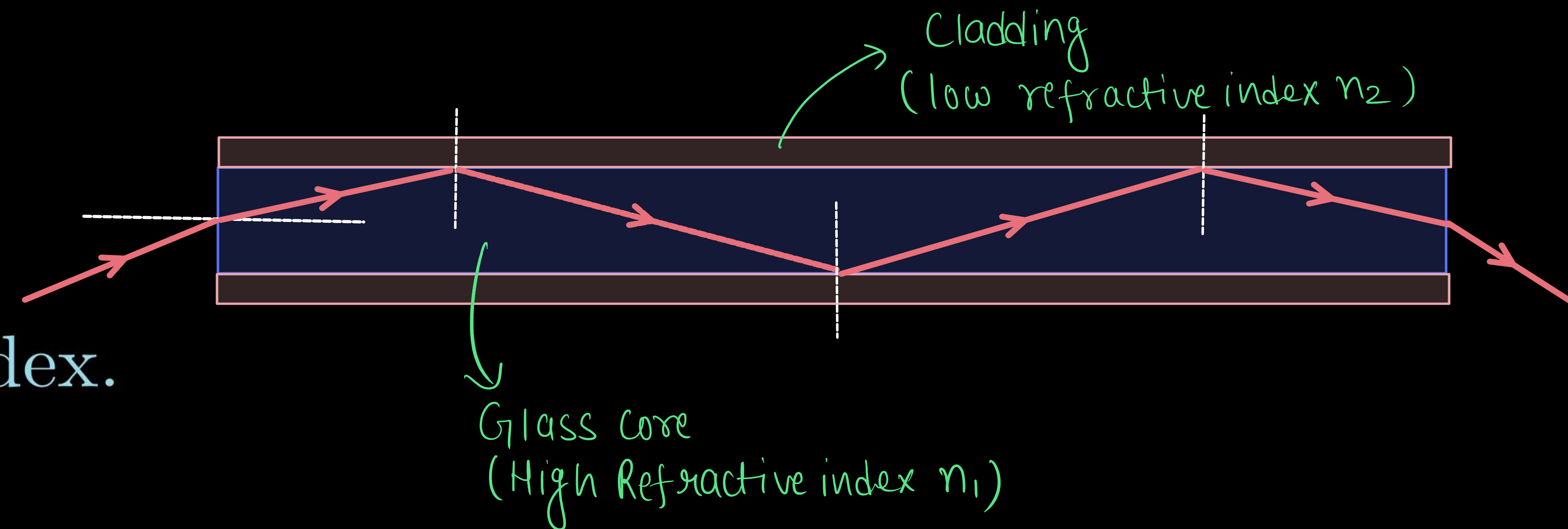


Optical Fibres

- **Optical fibre:** A flexible strand of glass (quartz) surrounded by a coating of low refracting index, that transmits light by **multiple total internal reflections**.

Each fibre has two parts:

- **Core:** Higher refractive index.
- **Cladding:** Slightly lower refractive index.
- **Principle:** Total Internal Reflection.



Light entering the core at a small angle undergoes repeated TIR at the core-cladding boundary and travels with negligible loss.

Condition:

$$n_{\text{core}} > n_{\text{cladding}}$$

- **Applications:** In optical communication systems, medical endoscopy, and light guides.

A point source is placed at the bottom of a tank containing a transparent liquid (refractive index n) to a depth H . The area of the surface of the liquid through which light from the source can emerge out is

In $\triangle ABC$ -

$$\tan i_c = \frac{BC}{AB}$$

$$\frac{\sin i_c}{\cos i_c} = \frac{R}{H} \quad \text{--- (1)}$$

Also,

$$\sin i_c = \frac{1}{n}$$

And

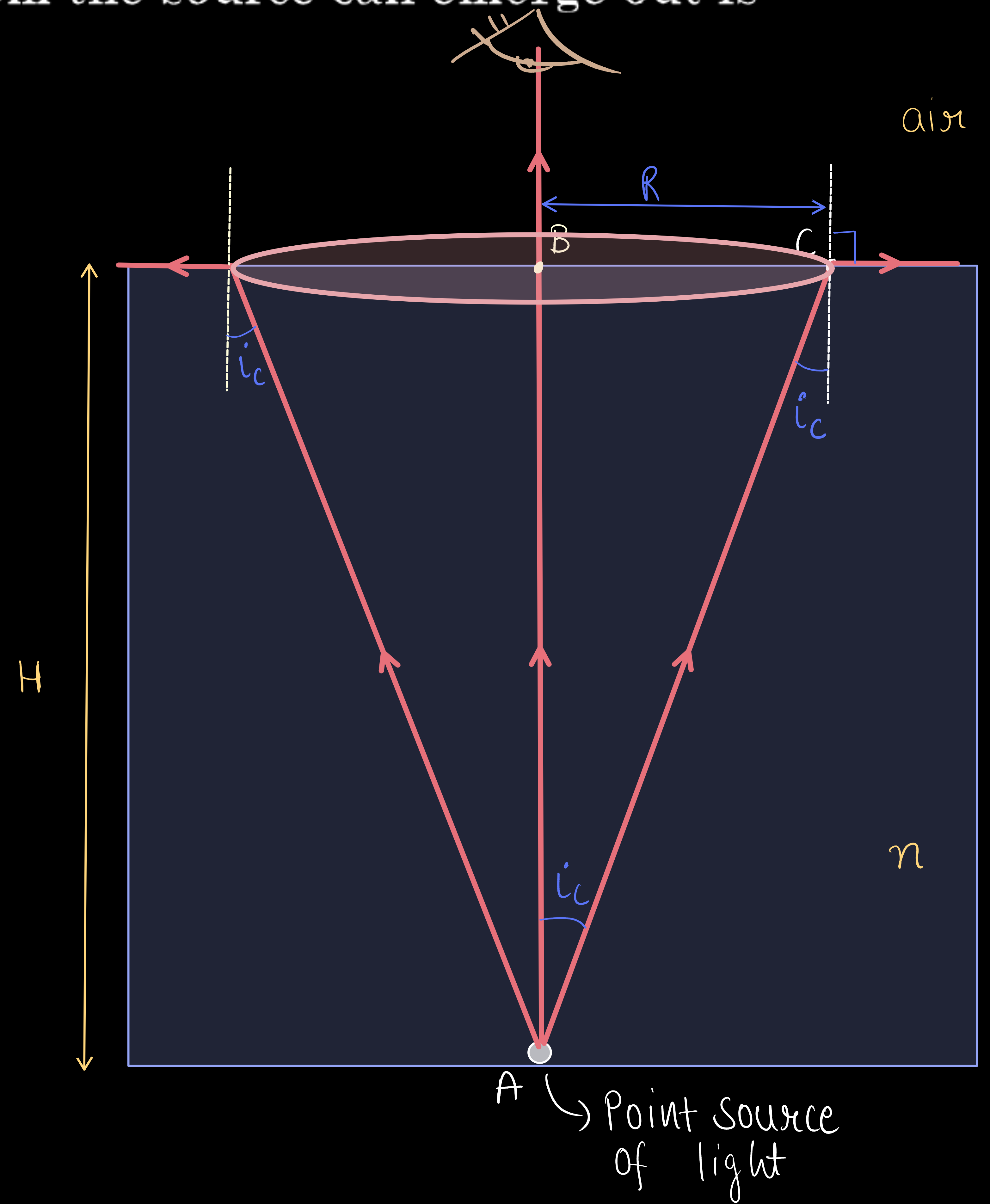
$$\cos i_c = \sqrt{1 - \sin^2 i_c} = \sqrt{1 - \frac{1}{n^2}}$$

$$\cos i_c = \frac{\sqrt{n^2 - 1}}{n}$$

$$\therefore \frac{\sin i_c}{\cos i_c} = \frac{1}{\sqrt{n^2 - 1}} \quad \text{--- (2)}$$

$\Rightarrow$ From eqⁿ (1) & (2)

$$R = \frac{H}{\sqrt{n^2 - 1}}$$



A point source is placed at the bottom of a tank containing a transparent liquid (refractive index n) to a depth H . The area of the surface of the liquid through which light from the source can emerge out is

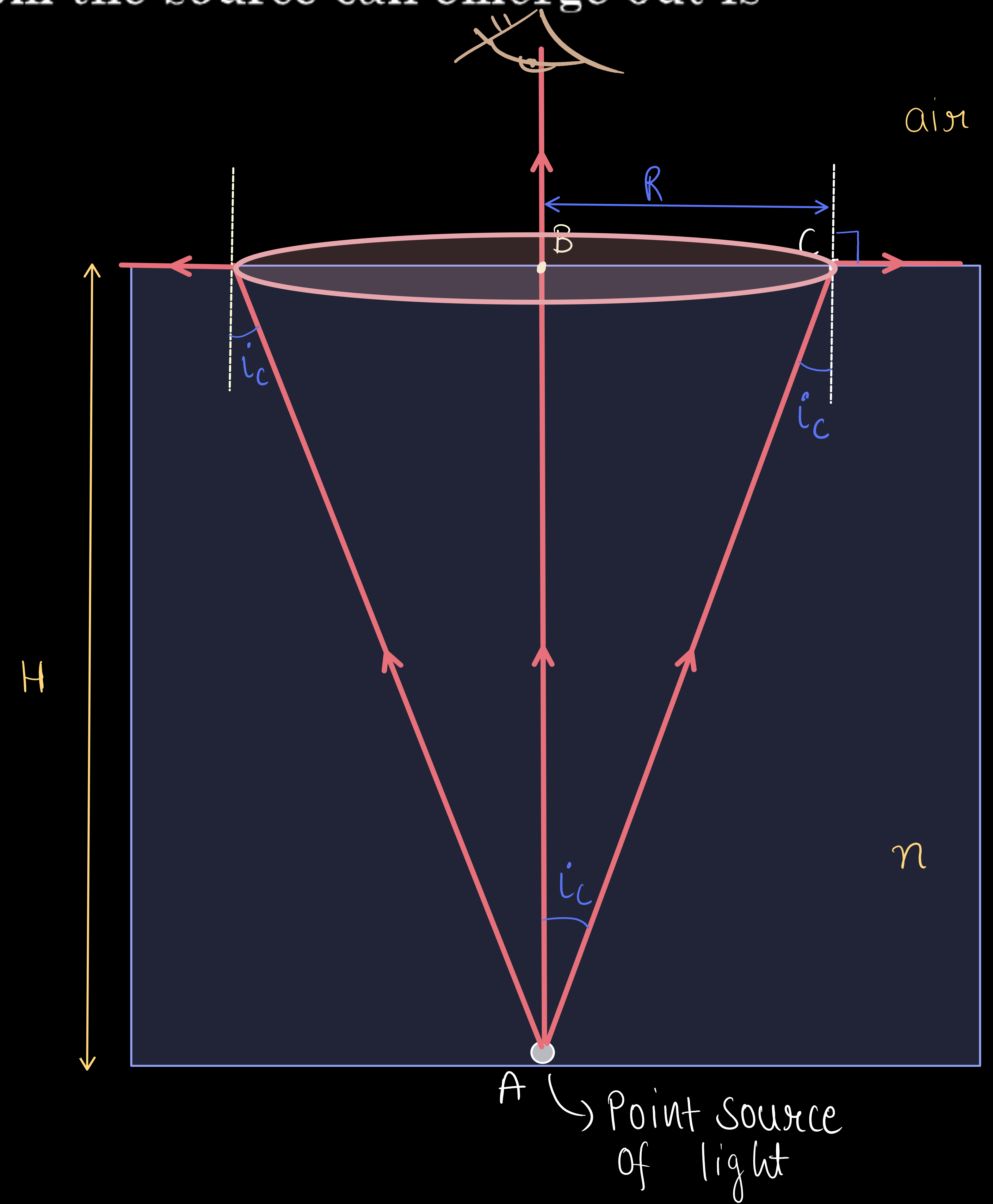
⇒ From eqⁿ (1) & (2)

$$R = \frac{H}{\sqrt{n^2 - 1}}$$

⇒ Area of the Circular disk -

$$A = \pi R^2$$

$$A = \frac{\pi H^2}{(n^2 - 1)}$$



Refraction of Light Through a Prism

- Triangular Glass Prism :

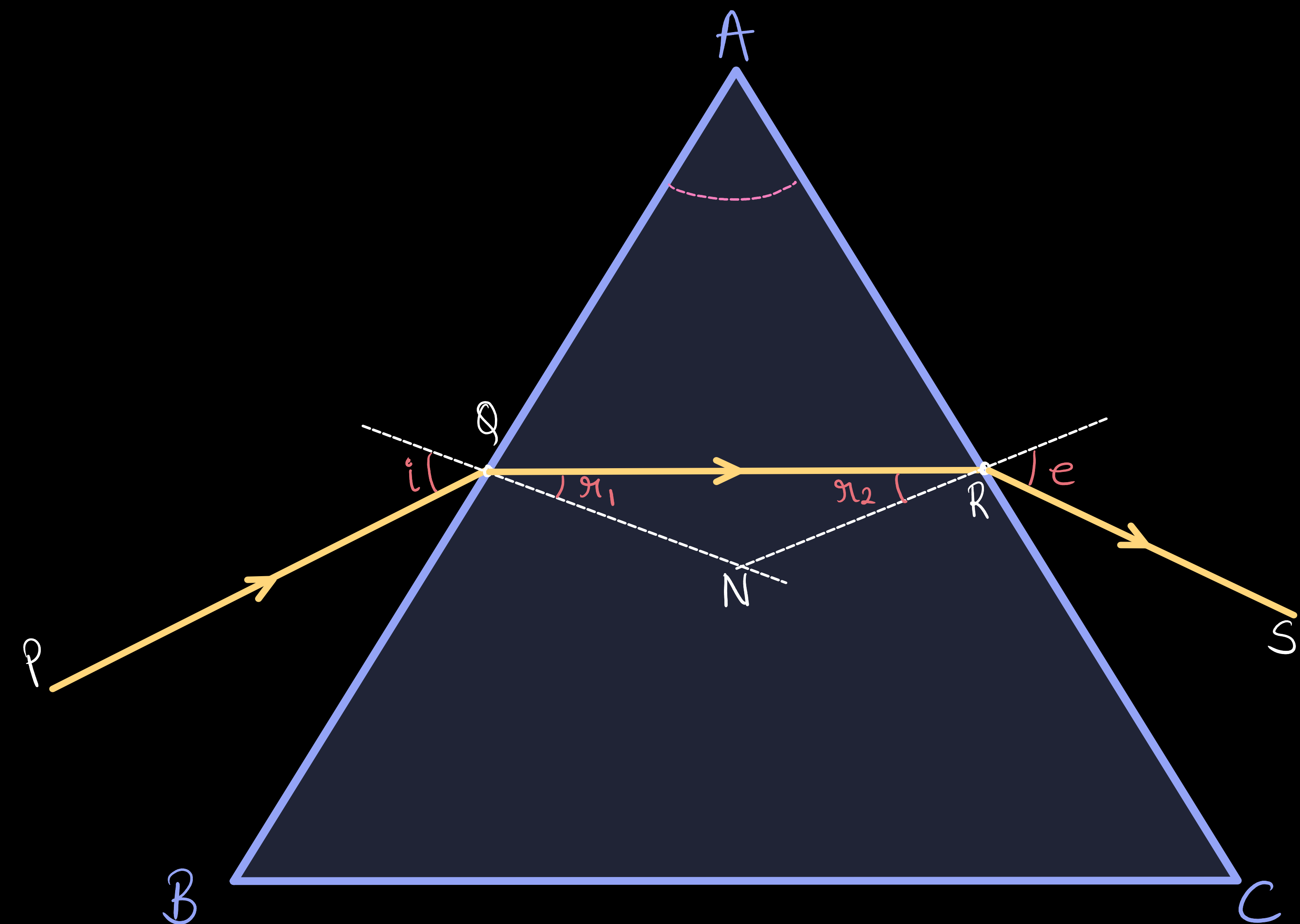
It has **two triangular bases** and **three rectangular lateral surfaces**. These surfaces are inclined to each other.

- Angle of the prism (A) :

The **angle between its two refracting faces** is called the angle of the prism.

- Refraction of light by prism :

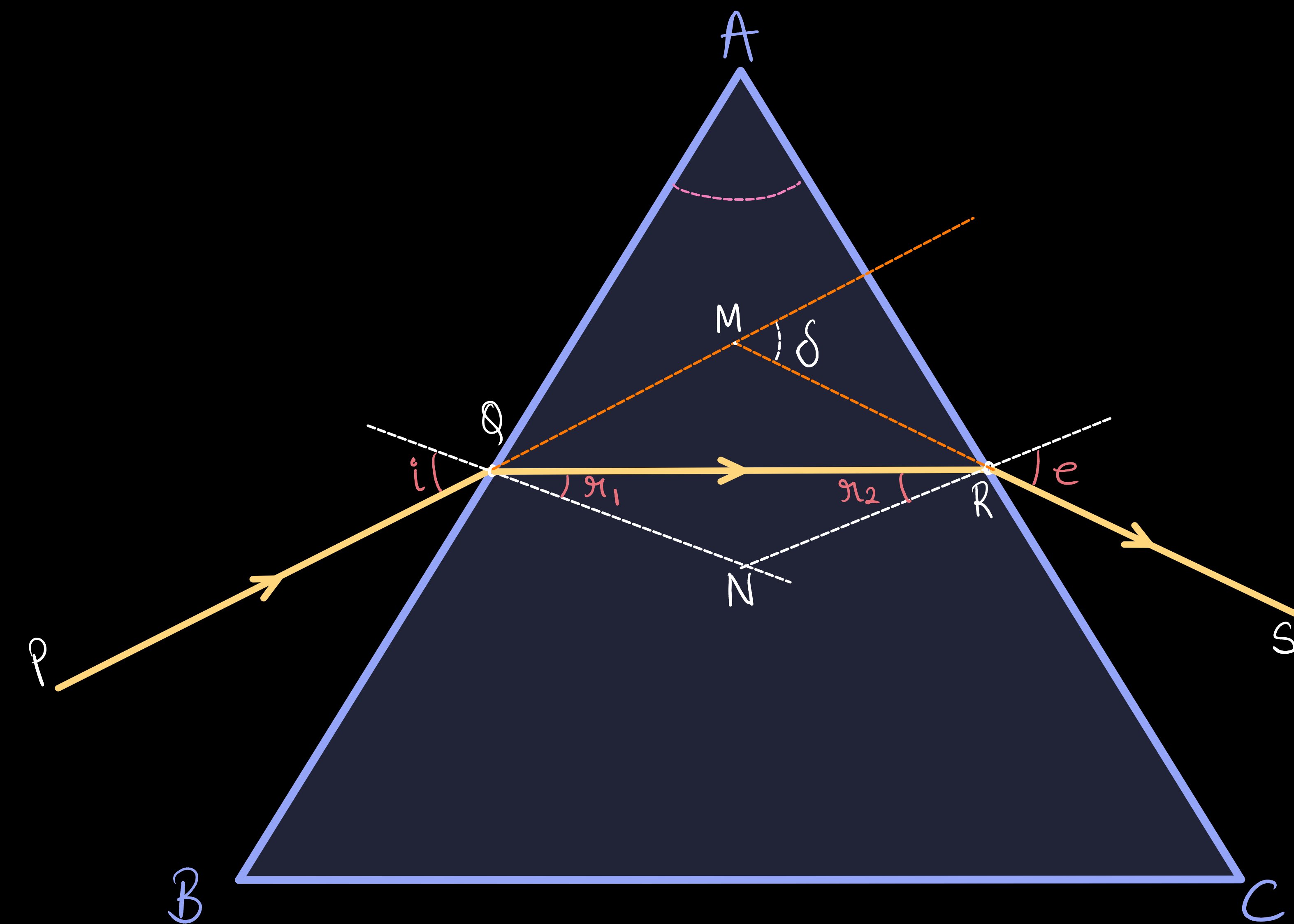
A ray of light is entering **from air to glass** at the **first surface AB**. The light ray on refraction has **bent towards the normal**. At the **second surface AC**, the light ray has **entered from glass to air**. Hence it has **bent away from normal**.



Refraction of Light Through a Prism

- Angle of Deviation (δ) :

The **peculiar shape of the prism** makes the emergent ray bend at an angle to the direction of the incident ray. This **angle between the direction of the incident ray and emergent ray** is called the angle of deviation.



Refraction Through a Prism (Derivation)

In ΔMQR -

$$\delta = i - r_1 + e - r_2 \quad (\text{external Angle property of } \Delta)$$

$$\delta = i + e - (r_1 + r_2) \quad - (1)$$

In ΔAQR -

$$A + 90 - r_1 + 90 - r_2 = 180^\circ \quad (\text{Angle sum property of } \Delta)$$

$$A - (r_1 + r_2) = 0$$

$$A = r_1 + r_2 \quad - (2)$$

From eqⁿ (1) & (2)

$$\delta = i + e - A \quad - (3)$$

At minimum deviation position -

$$\delta = \delta_m, r_1 = r_2 = r \quad \text{and} \quad i = e$$

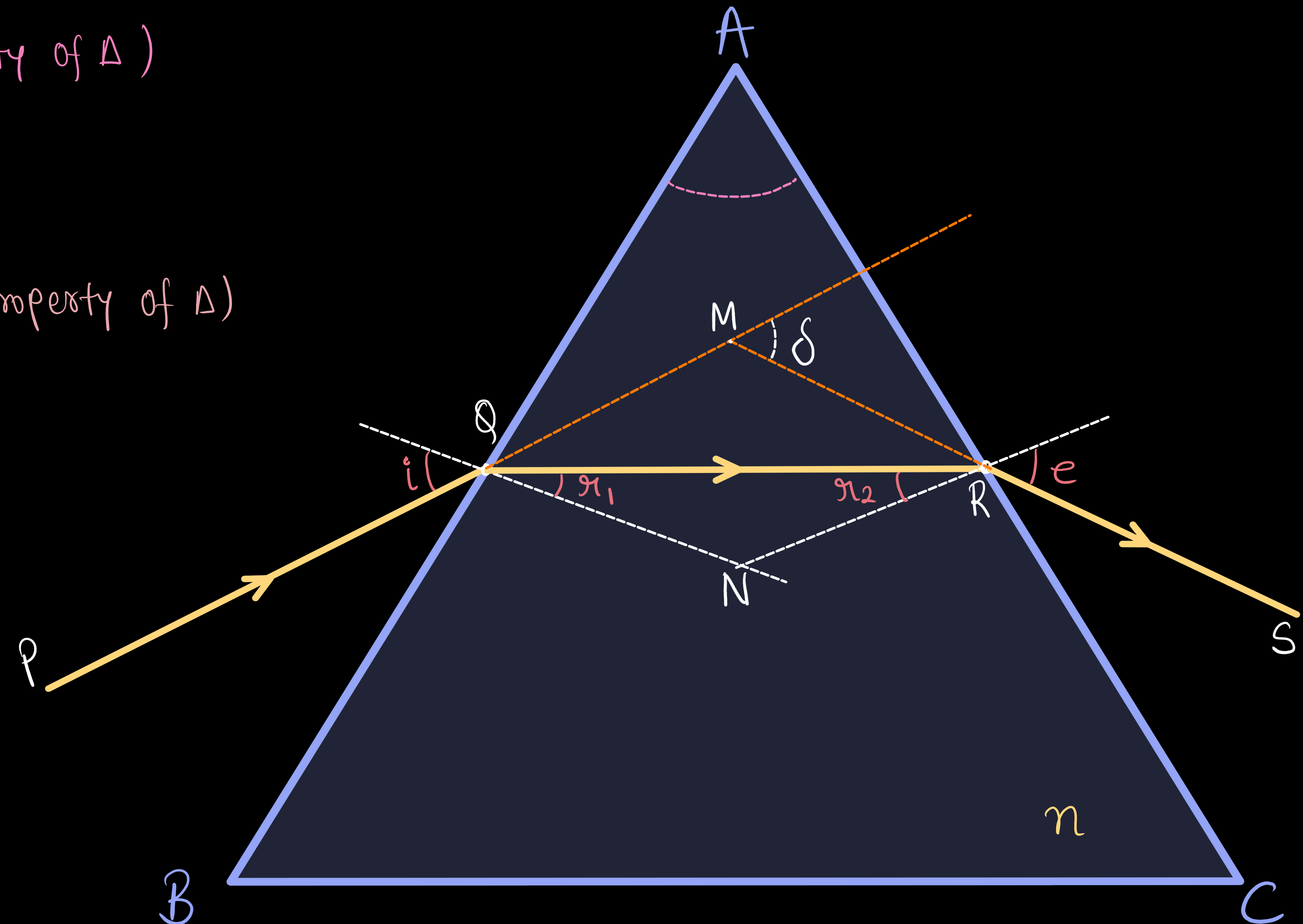
Also, the refracted ray (QR) is parallel to the base of the prism.

from eqⁿ (1) $A = r + r$
 $A = 2r$

$$r = \frac{A}{2} \quad - (4)$$

from eq(3) $\delta_m = i + i - A$
 $\delta_m = 2i - A$

$$\Rightarrow i = \frac{\delta_m + A}{2} \quad - (5)$$



Refraction Through a Prism (Derivation)

using Snell's at face AB -

$$1 \times \sin i = n \times \sin r$$

from eqⁿ (4) & (5)

$$i = \frac{A + \delta_m}{2} \quad \& \quad r = \frac{A}{2}$$

$$\therefore \sin\left(\frac{A + \delta_m}{2}\right) = n \sin\left(\frac{A}{2}\right)$$

$$n = \frac{\sin\left(\frac{A + \delta_m}{2}\right)}{\sin\left(\frac{A}{2}\right)}$$

★★

